

Autonomous Risk & Trust Standard (ARTS)

OriginID: OOF-OID-ASGA-ARTS-2026-07-06-0005

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Autonomous Systems Governance Architecture (ASGA™)

Operational Layer: Autonomous Operational Governance Layer

Category: Governance & Enforcement

Subcategory: Autonomous Operational Governance

Type: Parent Standard

Governed Space: Autonomous Operational Risk & Trust

Version: 1.0

Status: Canonical · Open Standard

Origin Date: 6 July 2026

Compatibility: OOF Methodology OS · Governance Architecture (GOA™) · Operational Reality Architecture (ORA™) · Accountability Governance Architecture (AGA™) · AI Governance Architecture (AIG®) · Cognitive Governance Intelligence Architecture (CLIA®) · Memory Governance Intelligence Architecture (MGIA™)

AI-Readable: Yes

Authority: OOF®

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL)

Governed Space

Autonomous Operational Risk & Trust

Core Question

What level of operational risk and trust is acceptable?

Canonical Definition

Autonomous Risk & Trust Standard (ARTS) defines the structural conditions under which autonomous operational risk is identified, evaluated, monitored, and governed while operational trust is established, preserved, measured, and continuously justified throughout the complete autonomous operational lifecycle.

ARTS governs autonomous operational risk and trust.

The standard establishes the governance conditions required to ensure that autonomous systems operate only within acceptable levels of operational risk while maintaining justified operational trust.

Autonomous execution creates operational outcomes.

Risk evaluates uncertainty.

Trust evaluates confidence.

ARTS governs both.

Governance Flow Position

Autonomous Identity → Autonomous Authority → Autonomous Execution → Autonomous Constraints → Autonomous Risk & Trust → Autonomous Coordination → Autonomous Escalation → Autonomous Operational Accountability → Autonomous Compatibility → Autonomous Evolution

Standard Abstract

Autonomous systems operate under varying operational conditions.

Some operations increase uncertainty.

Some operations increase confidence.

Governance requires both dimensions to remain continuously evaluated.

Without governed risk and trust:

- operational uncertainty increases,
- confidence declines,
- governance decisions weaken,
- autonomous operations become less predictable,
- operational legitimacy becomes difficult to justify.

ARTS establishes the governance conditions required to continuously balance operational risk with justified operational trust.

Operational Role

Autonomous Risk & Trust Standard provides the governance layer responsible for governing acceptable operational risk while preserving justified operational trust throughout autonomous operation.

Its role is to ensure that autonomous systems remain both operationally acceptable and governance-trustworthy.

Operational Space

ARTS governs:

- operational risk,
- operational trust,
- risk classification,
- trust evaluation,
- trust degradation,
- trust recovery,
- operational risk escalation,
- governance confidence.

This governed space ensures that autonomous operations remain continuously acceptable from both risk and trust perspectives.

Why This Standard Exists

Autonomous operation always involves uncertainty.

Governance must continuously determine whether operational risk remains acceptable and whether operational trust remains justified.

ARTS exists because autonomous operational risk and trust represent the fifth governable space of Autonomous Systems Governance Architecture (ASGA™).

Scope

This standard may be applied across:

- autonomous AI agents,
 - robotics,
 - autonomous vehicles,
 - industrial automation,
 - healthcare robotics,
 - enterprise AI platforms,
 - critical infrastructure,
 - defense autonomy,
 - Human-AI operational environments,
 - multi-agent ecosystems.
-

Relationship to Other OOF® Architectures

ARTS governs autonomous operational risk and trust.

It interoperates with:

- **GOA™** for governance principles,
 - **ORA™** for operational reality,
 - **AGA™** for accountability,
 - **AIG®** for AI behavioral governance,
 - **CLIA®** for cognition,
-

- **MGIA™** for memory governance.

Together these architectures support autonomous operations that remain governable, justifiable, and trustworthy under changing operational conditions.

Foundational Principle

Trustworthy autonomy is not achieved by eliminating risk.

Trustworthy autonomy is achieved by continuously governing acceptable operational risk while maintaining justified operational trust.

Canonical Closing Statement

Autonomous Risk & Trust establishes the governance conditions required to evaluate uncertainty, preserve confidence, and maintain trustworthy autonomous operation. By governing operational risk and operational trust together, ARTS strengthens resilient, accountable, and continuously governable autonomous systems.

Nahllad

Module Architecture

→ Operational Risk Classification Module (ORCM)

→ Operational Trust Evaluation Module (OTEM)

→ Trust Degradation Module (TDM)

→ Trust Recovery Module (TRM)

→ Operational Risk Escalation Module (OREM)

Related Documents

→ About Autonomous Risk & Trust Standard (ARTS)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>